

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company introduces automated code analysis tools to identify potential defects before coding errors occur in production. Which type of maintenance does this activity best support?

- A. Adaptive Maintenance
- B. Preventive Maintenance
- C. Perfective Maintenance
- D. Corrective Maintenance

Answer: B

2. Which of the following best describes the primary goal of Software Engineering?

- A. Writing programs with the shortest code.
- B. Developing high-quality software within budget and schedule.
- C. Reducing hardware manufacturing costs.
- D. Eliminating software maintenance.

Answer: B

3. Which Software Development Process Model is MOST appropriate when every completed phase must be formally verified before proceeding?

- A. Prototype Model
- B. Incremental Model
- C. V-Model
- D. Agile Model

Answer: C

4. A company wants customers to evaluate an early working version of a mobile application before final development begins. Which model is most appropriate?

- A. Waterfall Model
- B. Prototype Model
- C. V-Model
- D. RAD Model

Answer: B

5. Which requirement specifies that a system must recover automatically within 30 seconds after a server failure?

- A. Functional Requirement
- B. Business Requirement
- C. Reliability Requirement
- D. User Requirement

Answer: C

6. Which Requirements Engineering activity resolves inconsistencies between customer expectations and technical limitations?

- A. Validation
- B. Negotiation
- C. Inception
- D. Specification

Answer: B

7. Which document formally records agreed software requirements and serves as the foundation for future development?

- A. Test Plan
- B. User Manual
- C. Requirements Baseline
- D. Bug Report

Answer: C

8. Agile teams prefer customer feedback because it helps them:

- A. Freeze requirements permanently.
- B. Remove all project risks.
- C. Adapt to changing business needs.
- D. Avoid software testing.

Answer: C

9. Which human factor is considered essential for successful Agile software development?

- A. Individual competition
- B. Mutual trust and respect
- C. Strict organizational hierarchy
- D. Independent decision making without discussion

Answer: B

10. In XP, which artifact contains short descriptions of customer requirements?

- A. Sprint Backlog
- B. User Story Cards
- C. Design Specification
- D. Product Roadmap

Answer: B

11. In Extreme Programming (XP), what is the maximum recommended working week for developers?

- A. 30 Hours
- B. 35 Hours
- C. 40 Hours
- D. 60 Hours

Answer: C

12. Which Scrum role has the authority to organize its own work and determine how Sprint goals will be achieved?

- A. Product Owner
- B. Scrum Team
- C. Scrum Master
- D. Project Sponsor

Answer: B

13. Which Scrum meeting focuses primarily on planning work for the next Sprint?

- A. Sprint Planning Meeting
- B. Sprint Review Meeting
- C. Daily Scrum
- D. Sprint Retrospective

Answer: A

14. Which deliverable is produced during the Feasibility Study stage of DSDM?

- A. Functional Prototype
- B. Feasibility Report
- C. User Documentation
- D. Test Record

Answer: B

15. According to the MoSCoW prioritization technique, which category represents requirements that are optional but desirable?

- A. Must Have
- B. Should Have
- C. Could Have
- D. Won't Have

Answer: C

16. Which FDD process forms feature teams to perform detailed analysis and design?

- A. Build by Feature
- B. Design by Feature
- C. Develop Overall Model
- D. Plan by Feature

Answer: B

17. Which FDD role is primarily responsible for maintaining consistent terminology across the development team?

- A. Tester
- B. Toolsmith
- C. Language Guru
- D. Technical Writer

Answer: C

18. Which software myth assumes that the working software is the only deliverable required for a successful project?

- A. Customer Myth
- B. Professional Myth
- C. Management Myth
- D. Testing Myth

Answer: B

19. According to software failure studies, the majority of software faults originate from:

- A. Deployment Activities
- B. Coding Errors Only
- C. Specification and Design
- D. Hardware Failures

Answer: C

20. Which software engineering practice is intended to reduce defect correction costs by identifying problems during requirements and design?

- A. Shift-Left Testing
- B. Pair Programming
- C. Sprint Review
- D. Continuous Deployment

Answer: A